

Software Requirements Specification (SRS)

Smart EV Charging Station Management System

1. Introduction

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of the **Smart EV Charging Station Management System**. This system provides an online platform for electric vehicle users to locate charging stations, book charging slots, and manage bookings efficiently.

1.2 Scope

The Smart EV Charging Station Management System is a web-based application that enables EV users to:

- Register and log in to the system
- Search for available charging stations
- View charging slot availability
- Book charging slots
- Generate QR codes for booking verification
- Cancel or manage bookings

The system also allows administrators and station staff to manage charging stations and monitor bookings.

1.3 Definitions and Abbreviations

Term	Meaning
EV	Electric Vehicle
SRS	Software Requirements Specification
API	Application Programming Interface
DB	Database
JWT	JSON Web Token
QR	Quick Response Code

2. Overall Description

2.1 Product Perspective

The Smart EV Charging Station Management System is a **web-based application** that follows a **client-server architecture**.

The system includes:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js with Express.js
- **Database:** MySQL
- **Authentication:** JWT
- **Additional Feature:** QR Code generation for booking verification

2.2 Product Functions

The main functions of the system include:

- User registration and authentication
- Viewing charging stations
- Viewing available charging slots
- Booking charging slots
- Generating QR code for booking

- Viewing booking history
- Cancelling bookings
- Admin monitoring of stations and bookings

2.3 User Classes and Characteristics

EV User

- Electric vehicle owner
- Registers and books charging slots

Admin

- Manages charging stations
- Monitors all bookings

Station Staff

- Manages slot availability
- Verifies QR codes during charging

2.4 Operating Environment

Component Technology

Frontend HTML, CSS, JavaScript

Backend Node.js, Express.js

Database MySQL

Browser Chrome, Firefox, Edge

2.5 Design Constraints

- The system must use MySQL for data storage
- Authentication must be implemented using JWT
- The system must support web browsers

3. System Features

3.1 User Registration

Description

Allows users to create a new account.

Inputs

- Name
- Email
- Password
- Phone number

Output

User account successfully created.

3.2 User Login

Description

Users log in using email and password.

Input

- Email
- Password

Output

- Authentication token generated
- User logged in successfully

3.3 View Charging Stations

Description

Displays a list of available charging stations.

Input

User request to view stations

Output

List of stations with location and availability

3.4 View Available Slots**Description**

Displays available charging slots for a selected station.

Input

Station ID

Output

List of available time slots

3.5 Book Charging Slot**Description**

Allows users to reserve a charging slot.

Input

Slot ID

Output

Booking confirmation with QR code

3.6 Cancel Booking**Description**

Users can cancel previously booked slots.

Input

Booking ID

Output

Booking cancelled successfully

3.7 View Booking History**Description**

Displays all bookings made by the user.

Output

List of bookings with status and QR code

4. External Interface Requirements**4.1 User Interface**

The system provides a web interface with pages for:

- Registration
- Login
- Station listing
- Slot booking
- Booking history

4.2 Hardware Interface

Users access the system using a computer or mobile device connected to the internet.

4.3 Software Interface

The system interacts with:

- MySQL Database
- Web Browser
- QR Code generation library

5. Non-Functional Requirements

5.1 Performance Requirements

The system should respond to user requests within **2 seconds** under normal conditions.

5.2 Security Requirements

- User authentication using JWT
- Passwords stored securely
- Unauthorized access prevented

5.3 Reliability

The system should maintain data consistency during booking operations.

5.4 Usability

The system interface must be simple and easy to use.

5.5 Scalability

The system should support multiple users simultaneously.

6. System Architecture

The system follows a **client-server architecture**.

Components include:

1. Client Browser
2. Frontend Interface
3. Backend Server
4. REST API
5. Business Logic Modules
6. MySQL Database

Data flows from user interaction to backend services and database operations.

7. Database Requirements

The system database includes the following tables:

- Users
- Stations
- Slots
- Bookings

Relationships:

- A user can make multiple bookings
- A station can have multiple slots
- A booking references a user, station, and slot

8. Agile Development Approach

The project follows an **Agile development methodology** with iterative implementation.

Sprint Plan

Sprint Module

Sprint 1 Authentication Module

Sprint 2 Station Management

Sprint Module

Sprint 3 Slot Management

Sprint 4 Booking System

Sprint 5 QR Code Integration

Sprint 6 Frontend Integration and Testing

9. Assumptions and Dependencies

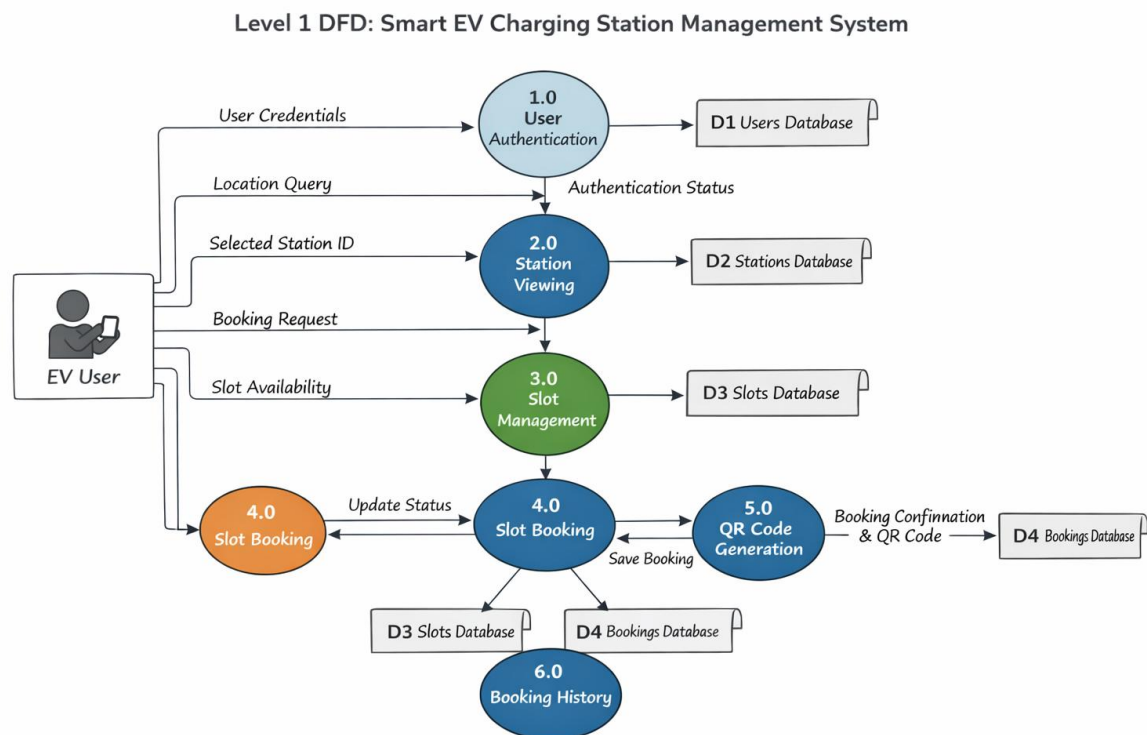
- Users have internet access
- Charging stations maintain accurate slot availability
- Database server is available continuously

10. Future Enhancements

Possible future improvements include:

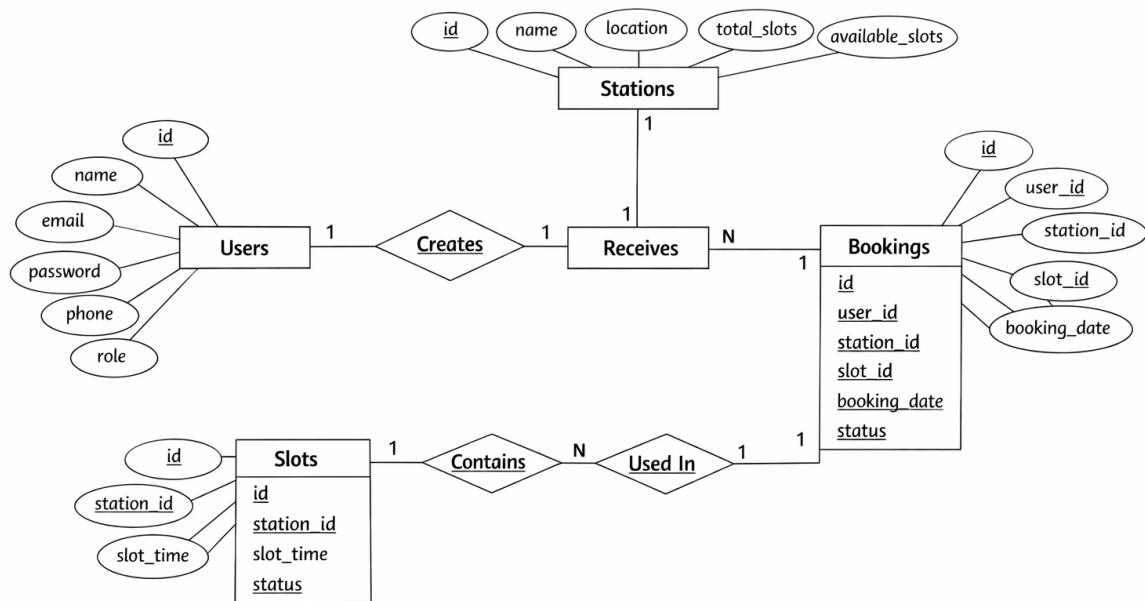
- Mobile application support
- Real-time slot availability tracking
- Payment integration
- Charging analytics dashboard

DATA FLOW DIAGRAM



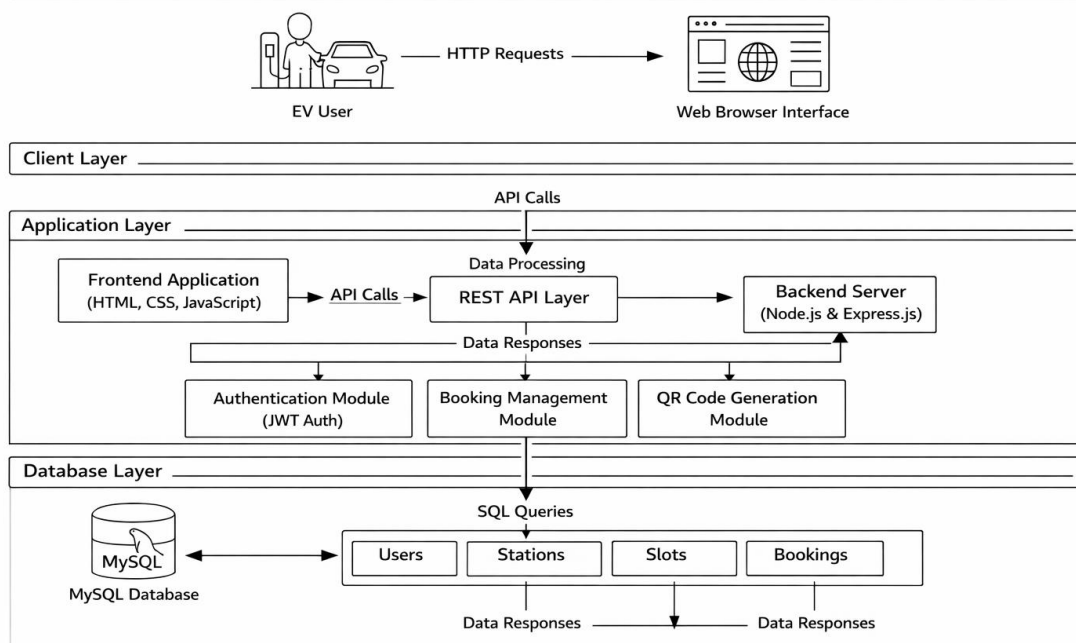
ER DIAGRAM

Entity Relationship Diagram – Smart EV Charging Station Management System

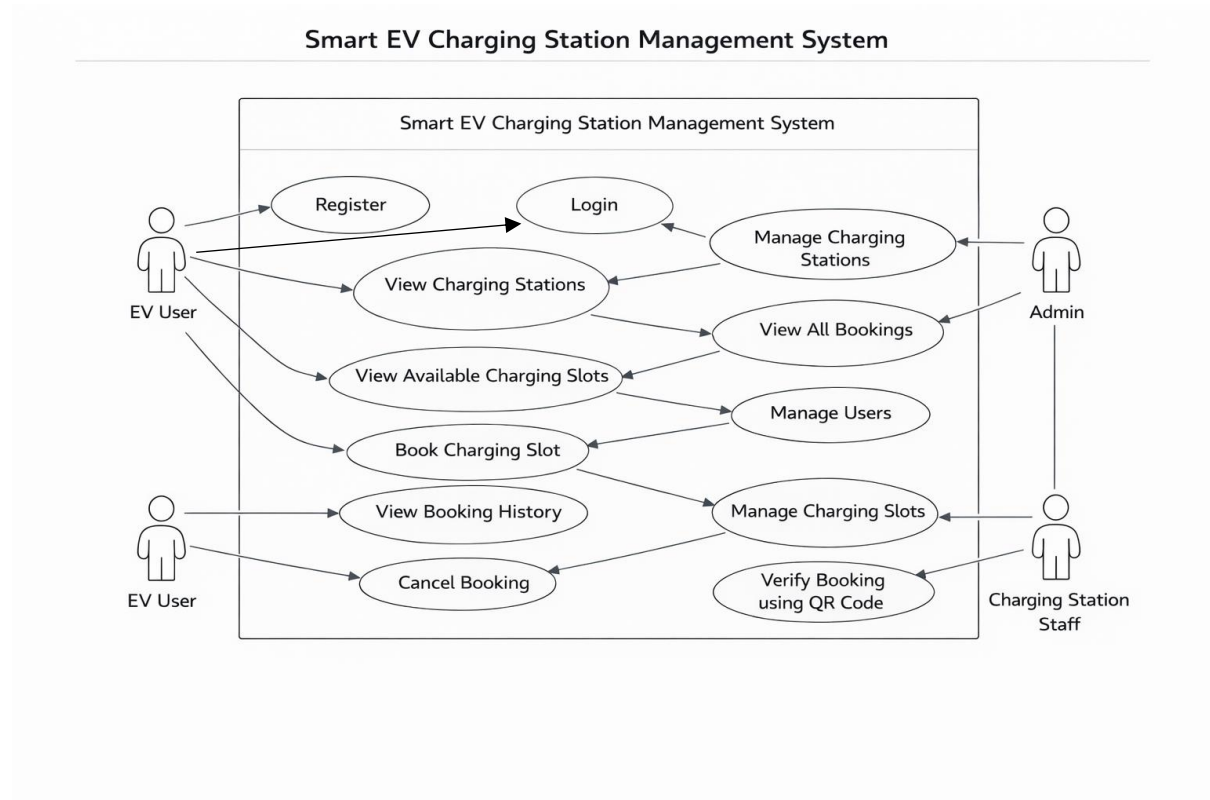


SYSTEM ARCHITECTURE DIAGRAM

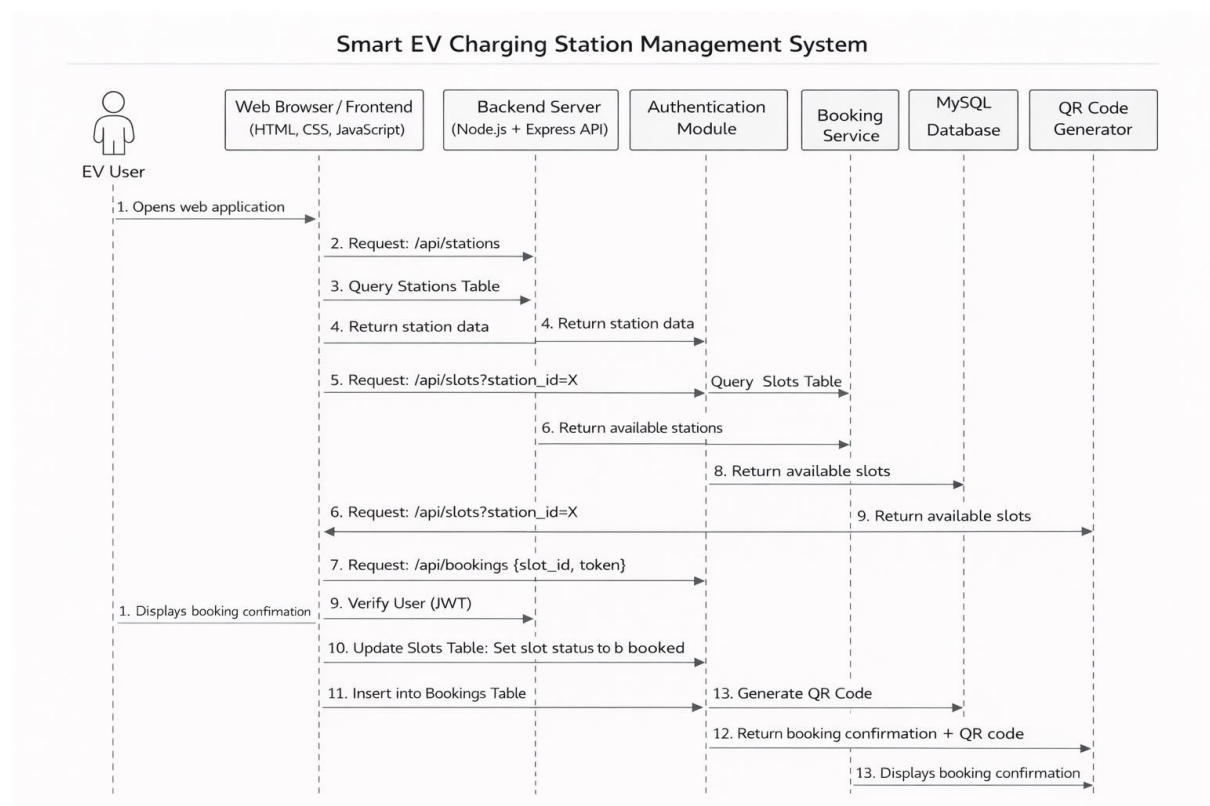
Smart EV Charging Station Management System



UML - USE CASE DIAGRAM



UML (SEQUENCE DIAGRAM)



Module-wise Description

The Smart EV Charging Station Management System is divided into several functional modules. Each module is responsible for a specific set of operations within the system.

-User Authentication Module

Description

This module handles user registration, login, and authentication. It ensures that only registered users can access the system's booking features.

Main Functions

- User Registration
- User Login
- Password validation
- JWT Token generation
- Secure session management

Technologies Used

- Node.js
- Express.js
- JWT Authentication
- MySQL database

Input

- User name
- Email
- Password
- Phone number

Output

- User account creation confirmation
- Authentication token for session access

- Charging Station Module

Description

This module manages information related to charging stations available in the system.

Main Functions

- View available charging stations
- Display station details
- Provide station location and slot availability

Data Stored

- Station ID
- Station name
- Location
- Available slots

Output

- A list of charging stations available to users.

- Slot Management Module

Description

This module manages charging time slots for each station.

Main Functions

- Create charging slots
- Display available slots
- Update slot availability
- Manage slot status (available/booked)

Data Stored

- Slot ID
- Station ID
- Slot time
- Slot status

Output

- List of available charging slots for selected stations.

6.4 Booking Module

Description

This module allows users to reserve charging slots and manage their bookings.

Main Functions

- Book charging slot
- Store booking details
- View booking history
- Cancel booking

Data Stored

- Booking ID
- User ID
- Station ID
- Slot ID
- Booking date
- Booking status

Output

- Booking confirmation and booking details.

- QR Code Generation Module

Description

This module generates QR codes for booking verification. The QR code contains booking information that can be verified by station staff.

Main Functions

- Generate QR code for each booking
- Encode booking information
- Display QR code to user

Technologies Used

- QR code generation library

Output

- QR code image associated with the booking.

- Admin Management Module

Description

This module allows administrators to monitor and manage system data.

Main Functions

- Manage charging stations
- Monitor user bookings
- Manage users
- View system activity

Output

- Administrative control panel for system monitoring.

Implementation Progress

The development of the Smart EV Charging Station Management System follows an Agile methodology with incremental implementation across multiple modules.

- Completed Modules

Module	Status	Description
User Registration	Completed	Users can create new accounts
User Login	Completed	Secure authentication using JWT
Charging Station Listing	Completed	Users can view available stations
Slot Viewing	Completed	Users can see available charging slots
Booking System	Completed	Users can reserve charging slots
QR Code Generation	Completed	QR code generated for booking verification
Booking History	Completed	Users can view their past bookings
Booking Cancellation	Completed	Users can cancel existing bookings

- Modules Under Development

Module	Status	Description
Admin Dashboard	In Progress	Interface for managing stations and bookings
Staff Module	In Progress	Station staff verification of bookings
System Analytics	Planned	Display charging usage statistics

- Development Milestones

The project development followed multiple iterative sprints.

Sprint Development Activity

Sprint 1 - Authentication module development

Sprint 2 - Charging station module implementation

Sprint 3 -Slot management system development

Sprint 4 -Booking system implementation

Sprint 5 -QR code integration

Sprint 6 -Frontend integration and testing

- Current System Status

- The system currently supports:
- User registration and login
- Viewing charging stations
- Viewing available charging slots
- Booking charging slots
- QR code generation for bookings
- Viewing booking history
- Cancelling bookings

The backend APIs are functional and integrated with the MySQL database, and the frontend interface allows users to interact with the system through a web browser.

Testing Plan

The testing plan for the **Smart EV Charging Station Management System** ensures that all modules of the system work correctly and meet the specified functional requirements. Testing focuses on verifying the core functionalities such as user authentication, station viewing, slot management, booking operations, and QR code generation.

The testing process includes the following stages:

- **Unit Testing:** Each module (authentication, station management, slot management, booking system) is tested individually.

- **Integration Testing:** Ensures proper interaction between frontend, backend APIs, and the MySQL database.
- **System Testing:** Tests the entire system workflow including user registration, login, slot booking, QR code generation, and booking cancellation.
- **User Acceptance Testing:** Confirms that the system meets user requirements and provides the expected functionality.

The testing environment includes a web browser (Google Chrome/Edge), Node.js backend server, MySQL database, and the frontend interface built using HTML, CSS, and JavaScript.

Test Cases

Test Case ID	Test Scenario	Input	Expected Result
TC01	User Registration	Valid user details (name, email, password, phone)	User account is successfully created
TC02	User Login	Correct email and password	User login successful and authentication token generated
TC03	Invalid Login	Incorrect password	Error message displayed
TC04	View Charging Stations	Request station list	List of charging stations displayed
TC05	View Available Slots	Station ID	Available charging slots displayed
TC06	Book Charging Slot	Valid slot ID	Slot booking created successfully
TC07	QR Code Generation	Successful booking	QR code generated and displayed
TC08	View Booking History	Logged-in user request	List of user bookings displayed
TC09	Cancel Booking	Valid booking ID	Booking status updated to cancelled
TC10	Unauthorized Access	Access system without login	Access denied message displayed